Graphing Proportional Relationships
Answer Key

Grades 6–7 Mathematics

Quick Answers

1. 3
2. 2.5
3. 4
4. (b) unit rate = 2.5, —
5. 1
6. (a) the equation = 6, (b) distance after 7 seconds = 42
7. (b) ratio at x = 2 = 4, (b) ratio at x = 3 = $\frac{11}{3}$, —
8. (a) the equation = 8.5, (b) shirts for 51 dollars = 6
9. (a) missing y-value = 17.5, (a) missing x-value = 2, —
10. (a) rate comparison = <, (b) cost of 12 cans at the cheaper store = 30

What is verified

7 of 10 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 4, 7, 9. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 6–7 Mathematics
7.RP.A.2 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–4 of 5 across 18 tagged checks.

1. Hose filling a tank: unit rate in liters per minute from the graph.
Solution: The line passes through the origin and the marked point $(2, 6)$. Unit rate $= \dfrac{\text{rise}}{\text{run}} = \dfrac{6-0}{2-0} = \dfrac{6}{2}$, so the tank fills at $\boxed{3}$ liters per minute. (Check: at $x = 1$ the line is at $y = 3$.)

2. Grapes: price per pound from the graph.
Solution: The line passes through $(0, 0)$ and the marked point $(4, 10)$. Price per pound $= \dfrac{10-0}{4-0} = \dfrac{10}{4}$, giving $\boxed{2.5}$ dollars per pound.

3. Write the equation $y = kx$ from the graph.
Solution: The marked point is $(5, 20)$, so $k = \dfrac{20-0}{5-0} = 4$. The equation is $\boxed{y = 4x}$

4. Muffin recipe table: plot the points; find the unit rate.
Solution (a): *(instructor-judged)* The points $(0,0)$, $(2,5)$, $(4,10)$, $(6,15)$ all lie on one straight line through the origin — the graph should show them joined by that line.
Solution (b): Using $(2,5)$ and $(6,15)$: rate $= \frac{15-5}{6-2} = \frac{10}{4}$, so $\boxed{2.5}$ cups per batch.

5. Two runners: how much greater is A's unit rate?
Solution: Runner A: $\frac{12}{2} = 6$ meters per second. Runner B: $\frac{10}{2} = 5$ meters per second. Difference: $6 - 5 = 1$, so A's rate is greater by $\boxed{1}$ meter per second.

6. Remote-control car: equation, then distance at 7 seconds.
Solution (a): The marked point is $(3,18)$, so $k = \frac{18}{3} = 6$ and the equation is $\boxed{y = 6x}$
Solution (b): Substitute $x = 7$: $y = 6 \cdot 7$, so the car travels $\boxed{42}$ meters.

7. Raffle tickets: plot, compute both ratios, decide proportionality.
Solution (a): *(instructor-judged)* Points $(1,4)$, $(2,8)$, $(3,11)$ plotted; the first two line up with the origin but the third does not.
Solution (b): At $x = 2$: $\frac{8}{2}$, so $\boxed{4}$ dollars per ticket. At $x = 3$: $\frac{11}{3}$, so $\boxed{\frac{11}{3}}$ dollars per ticket (about \$3.67).
Solution (c): *(instructor-judged)* Not proportional: the cost per ticket changes, so the three points do not lie on one straight line through the origin. Full credit for either the equal-ratios test or the straight-line-through-origin test.

8. Shirt printing: equation, then shirts for \$51.
Solution (a): The marked point is $(2,17)$, so $k = \frac{17}{2} = 8.5$ and the equation is $\boxed{y = 8.5x}$
Solution (b): Solve $8.5x = 51$: $x = \frac{51}{8.5}$, so you can buy $\boxed{6}$ shirts.

9. Complete the proportional table, then graph it.
Solution (a): The unit rate is $\frac{3.5}{1} = 3.5$. Missing x-value: solve $3.5x = 7$, giving $\boxed{2}$
Missing y-value: $y = 3.5 \cdot 5$, giving $\boxed{17.5}$
Solution (b): *(instructor-judged)* The completed points $(1,3.5)$, $(2,7)$, $(5,17.5)$ lie on one straight line through the origin.

10. Store A (15 dollars for 6 cans, graphed) vs. Store B ($y = 2.75x$): compare rates; cost of 12 cans at the cheaper store.

Solution (a): Store A's rate: $\dfrac{15}{6} = 2.5$ dollars per can. Store B's rate: 2.75 dollars per can. So $\boxed{2.5 < 2.75}$ — Store A is cheaper.

Solution (b): At Store A: $\dfrac{15}{6} \cdot 12$, so 12 cans cost $\boxed{30}$ dollars.